

People's Democratic Republic of Algeria
Ministry of higher education and scientific research



Ecole supérieure en sciences et technologies de l'informatique et du numérique



Classification part 2

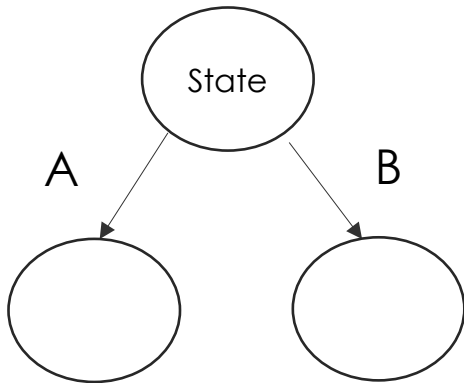
Decision Tree

- A prediction model presented in the form of a tree
- Each node of the tree tests a condition on a variable, and each child node corresponds to a label
- Originally, it applies to qualitative values
- Extended to take into consideration quantitative values

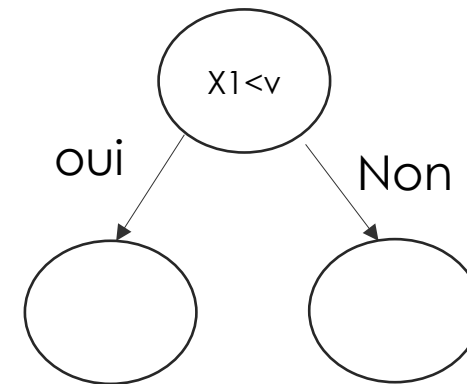
Decision Tree: Space partitioning

- A decision tree partitions the observation space X in a recursive manner until it reaches pure leaves (homogeneous regions) or a stopping criterion

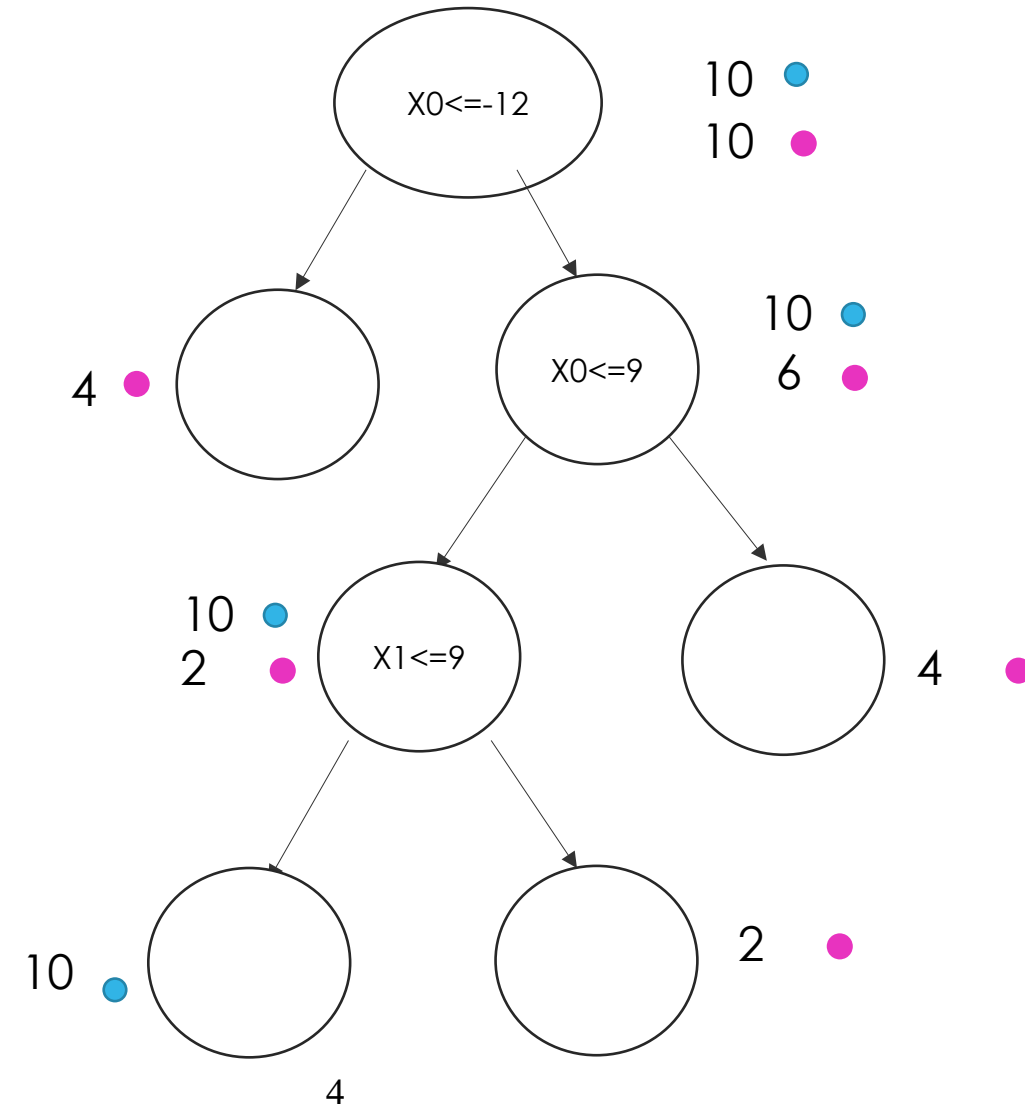
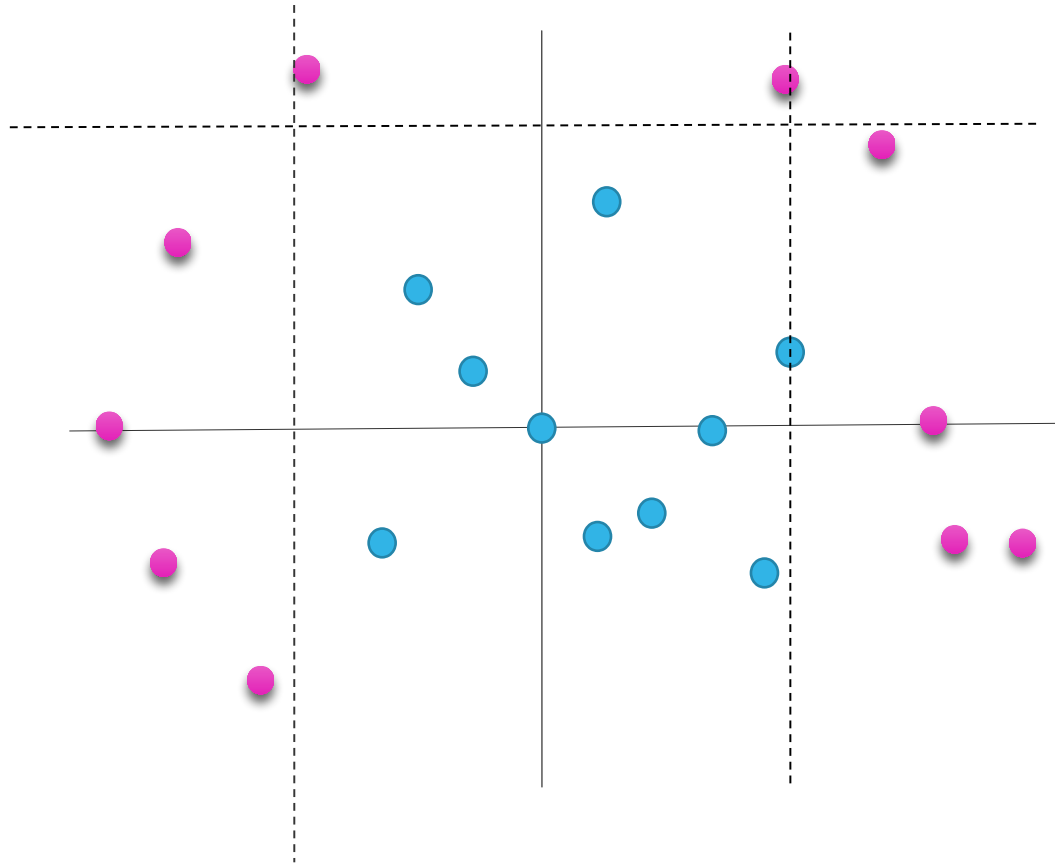
x_i is qualitative



x_i is quantitative



Decision Tree: Space partitioning



Decision Tree: Space partitioning

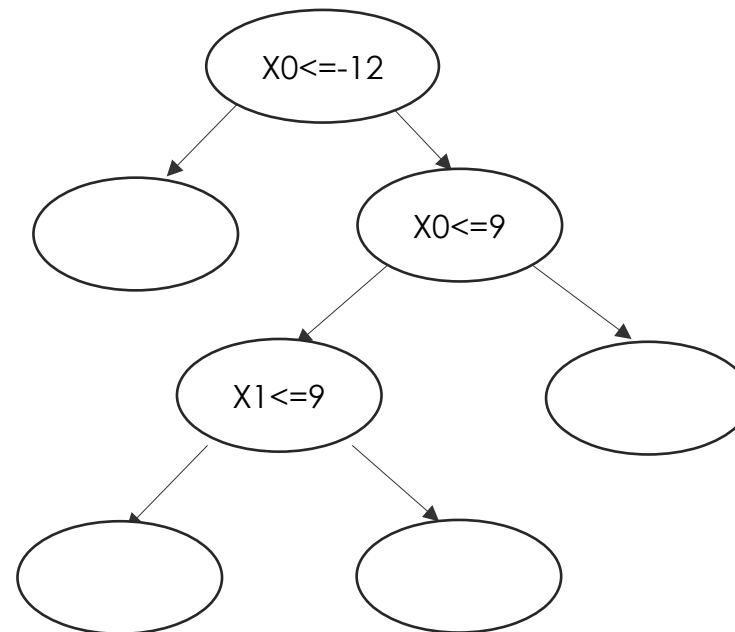
The decision tree contains two types of nodes :

Decision node: Contains a condition to split the data

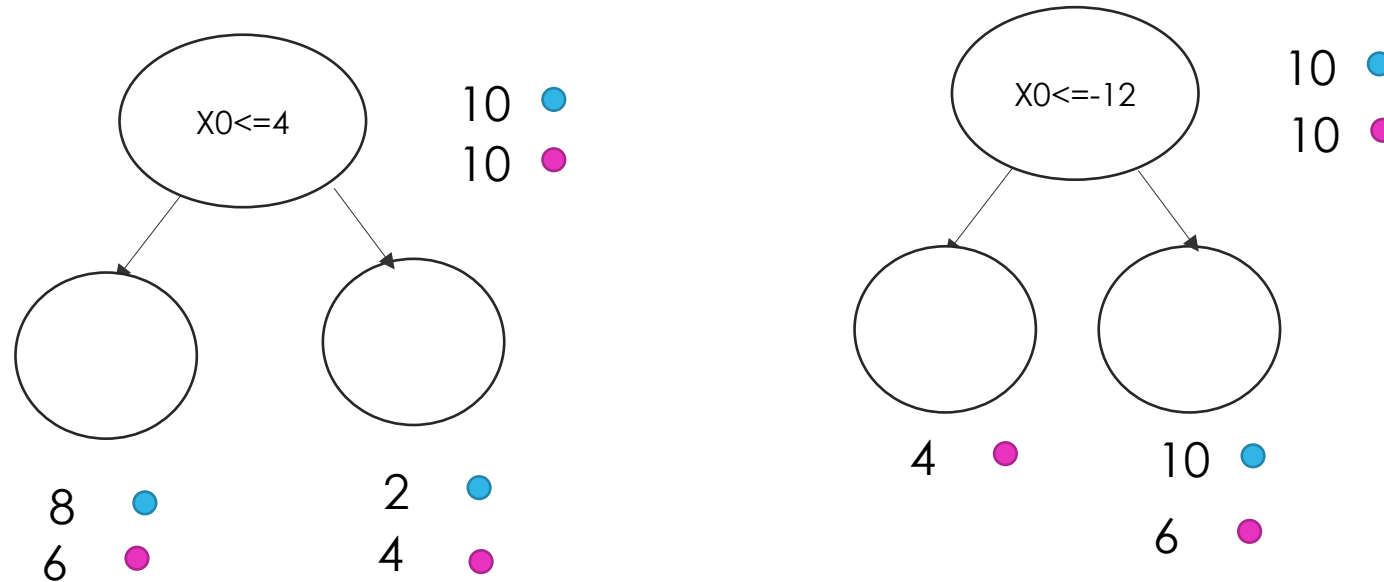
Leaf node or pure node: Contains only one class of data; it is used to determine the class of an object.

Decision Tree: Classification

- Let $x = (15, 7)$
- To classify X , we will traverse the tree until we reach a pure node.



Decision Tree: Learning Process



- Which is the best split?
- How does the model decide on the same split?

Decision Tree: Learning Process

We need to know what the correct conditions are to build our model in the form of a decision tree.

What features should be considered?

What value should be considered optimal?"

Where to Stop?

Decision Tree: Metrics

Different algorithms use different metrics for measuring the best

These generally measure the homogeneity of the target variable within the subsets

The model will consider the split that maximizes the information gain

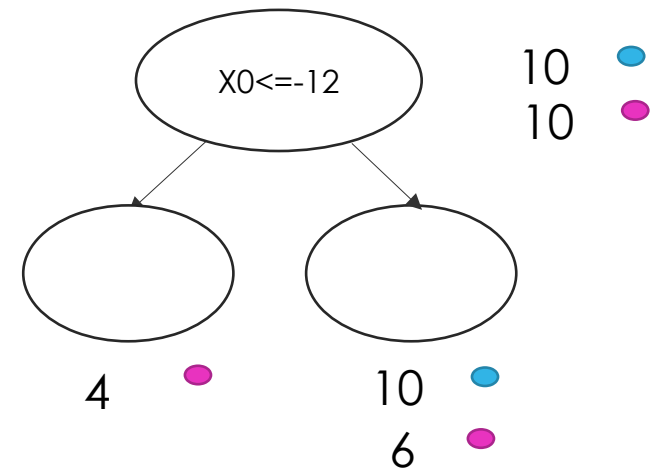
What is information gain?

Decision Tree: Information Gain

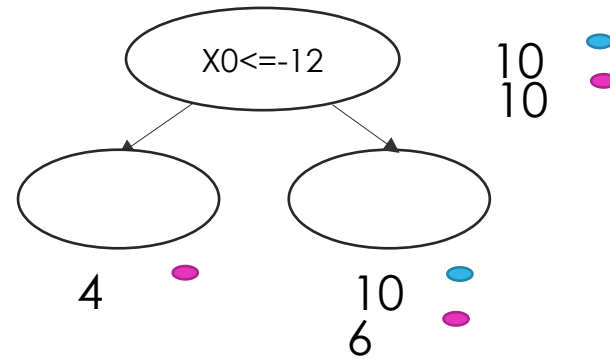
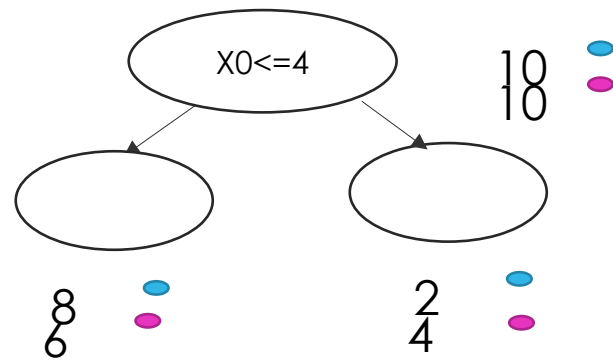
Information Gain ($IG(X, Y)$): It measures what information X provides about Y . (ID3 algorithm)

For example, the root node contains 50% pink class and 50% blue class, so falling into the correct class of an object is 50%, and it is the highest uncertain impurity."

is the relative size of a child node respecting the parent



Decision Tree: Information Gain



$$IG1 = 1 - \frac{14}{20} * 0,99 - \frac{6}{20} * 0,91 = 0,034$$

$$IG2 = 1 - \frac{4}{20} * 0 - \frac{16}{20} * 0,95 = 0,25$$

$IG2 > IG1$; therefore, we will choose the second split.

Decision Tree: Gini impurity

- Used by the CART“
- It measure incorrectly labeled of randomly chosen element. “
- Gini impurity can be computed by summing the probability of each item being chosen times the probability of a mistake in categorizing that item.
- It reaches its minimum (zero) when all cases in the node fall into a single target category.

Applicative example

Day	Outlook	Temp.	Humidity	Wind	Decision
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Decision Tree: Pruning

- is a technique used in decision tree algorithms to reduce the size of the tree by removing certain branches
- The goal is to prevent overfitting and improve the generalization of the model
- Pruning involves cutting off branches that do not provide significant additional predictive power, thereby simplifying the tree.

Decision Tree: Pruning

Pruning can be performed in different ways, such as:

- **pre-pruning:** branches are removed during the tree construction process
- **post-pruning:** the tree is initially grown to its full size and then unnecessary branches are pruned.

Decision Tree: Pre-Pruning

Pre-pruning, also known as "early stopping". The growth of the tree is halted based on certain criteria or conditions:

Maximum Depth Limit: Restricting the maximum depth of the tree. Once a certain depth is reached, further splitting is stopped.

Minimum Samples for Split: Setting a threshold for the minimum number of samples required to make a split. If a node has fewer samples than this threshold, it is not split further.

Decision Tree: Pre-Pruning

Maximum Leaf Nodes: Limiting the total number of terminal nodes (leaves) in the tree. Once this limit is reached, further splitting is prevented.

Minimum Impurity Decrease: Specifying a minimum improvement in impurity (e.g., Gini impurity or information gain) that a split must achieve. If a split doesn't meet this criterion, it is not performed.

Decision Tree: Post-Pruning

Build the complete model and then prune the elements contributing to overfitting.

Either replace a node with a leaf or replace two nodes with one.

1. At each non leaf node in the tree, calculate expected error rate if that sub tree is pruned
2. Calculate the expected error rate for that node if that tree is not pruned
3. if pruning node leads to greater expected error rate than keep the subtree, otherwise prune it

Post-Pruning: Minimum error pruning

$$W(T_k, d) = \frac{MC(d, k) - MCT(d, k)}{n(k) * (nt(d, k) - 1)}$$

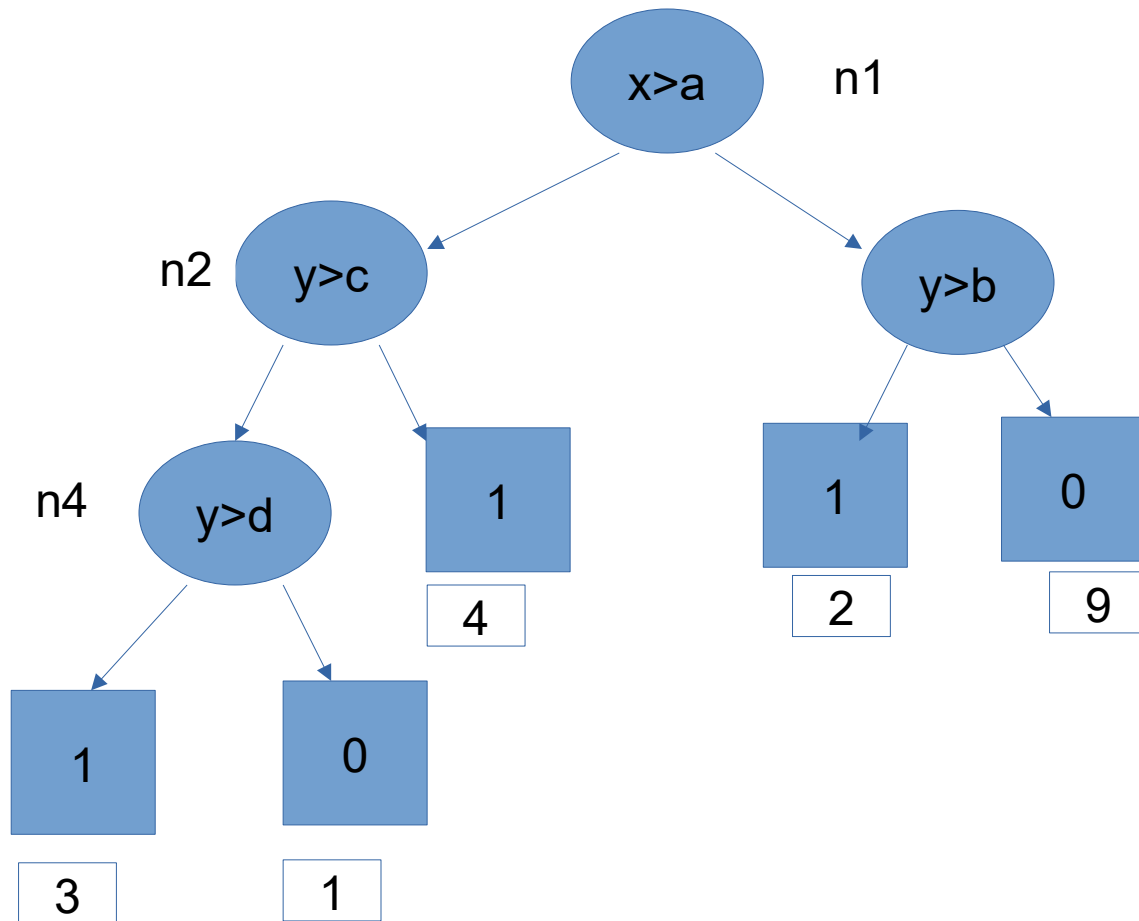
$MC(d, k)$: The number of individuals from the training set misclassified by node d of T_k when we assume it has been transformed into a leaf

$MCT(d, k)$: The number of individuals from the training set misclassified by the leaves of T_k located under node d

$n(k)$: The number of leaves in T_k

$nt(d, k)$: The number of leaves in the subtree of T_k located under node d

Decision Tree: Post-Pruning



Node n1:

0: 10

1: 9

$$w(T, n1) = \frac{9-0}{5(5-1)} = 0.45$$

n3 Node n2:

0: 1

1: 7

$$W(T, n2) = \frac{1-0}{5(3-1)} = 0.10$$

Node n3:

0: 9

1: 2

$$W(T, n3) = \frac{2-0}{5(2-1)} = 0.40$$

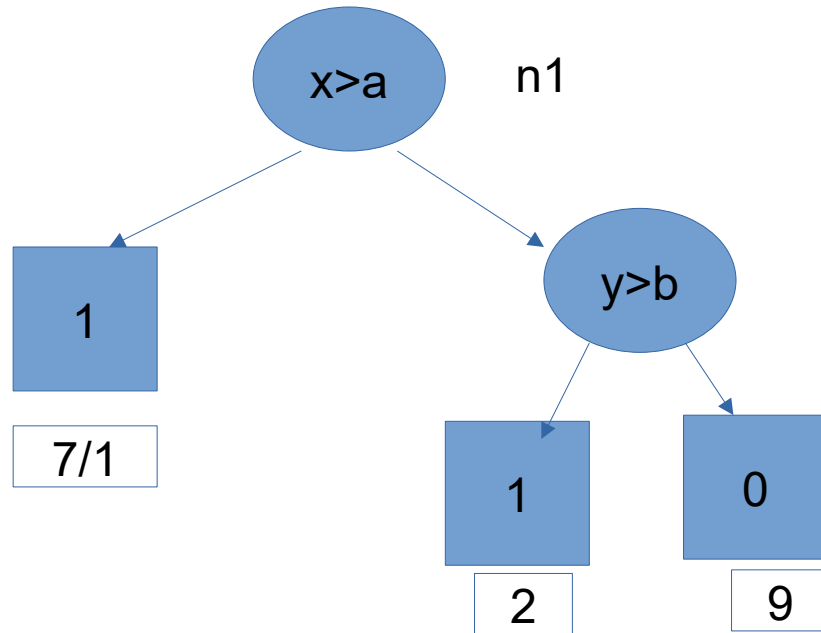
Node n4 :

0: 1

1: 3

$$W(T, n4) = \frac{1-0}{5(2-1)} = 0.20$$

Decision Tree: Post-Pruning



Node n1:

0: 10

1: 9

$$w(T, n1) = \frac{9-0}{5(5-1)} = 0.45$$

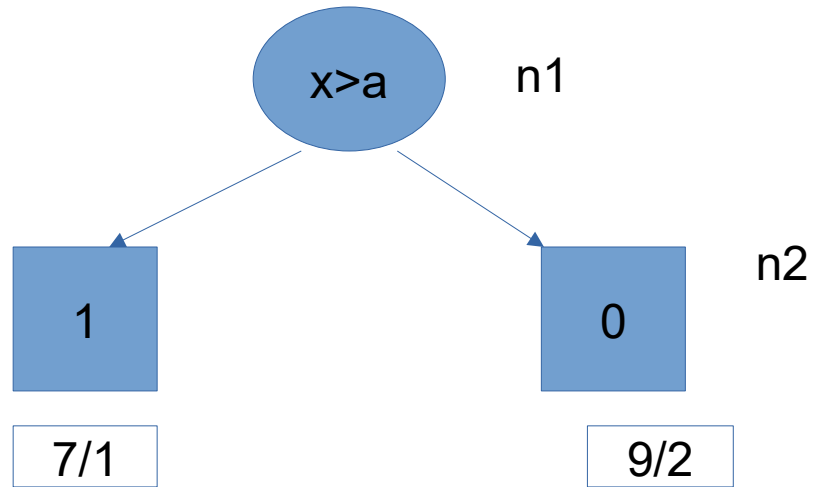
n2 Node n2:

0: 1

1: 7

$$W(T, n2) = \frac{2-0}{3(2-1)} = 0.66$$

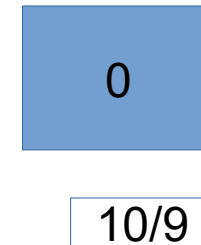
Decision Tree: Post-Pruning



Node n1:
0: 9
1: 7

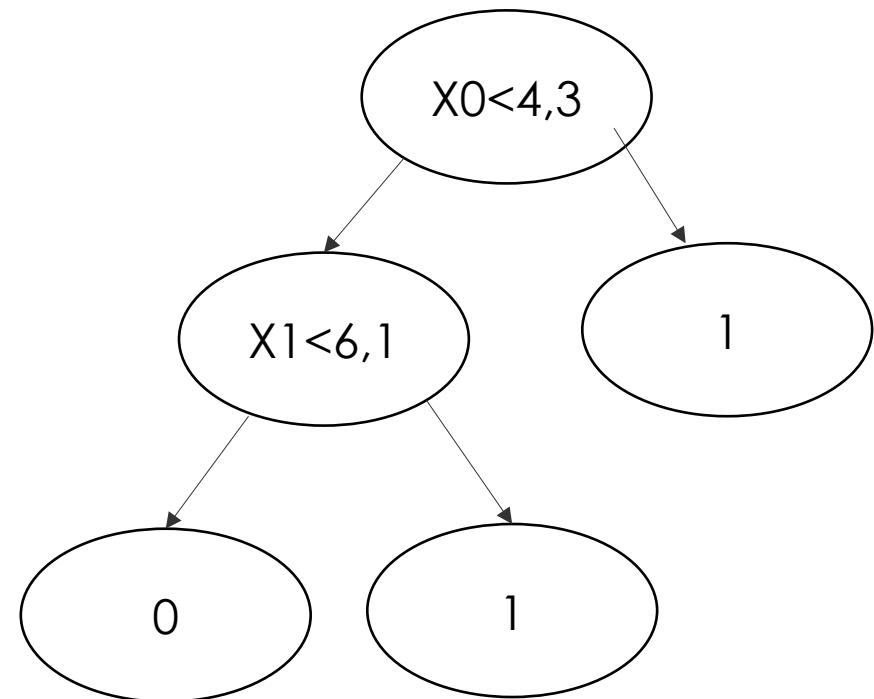
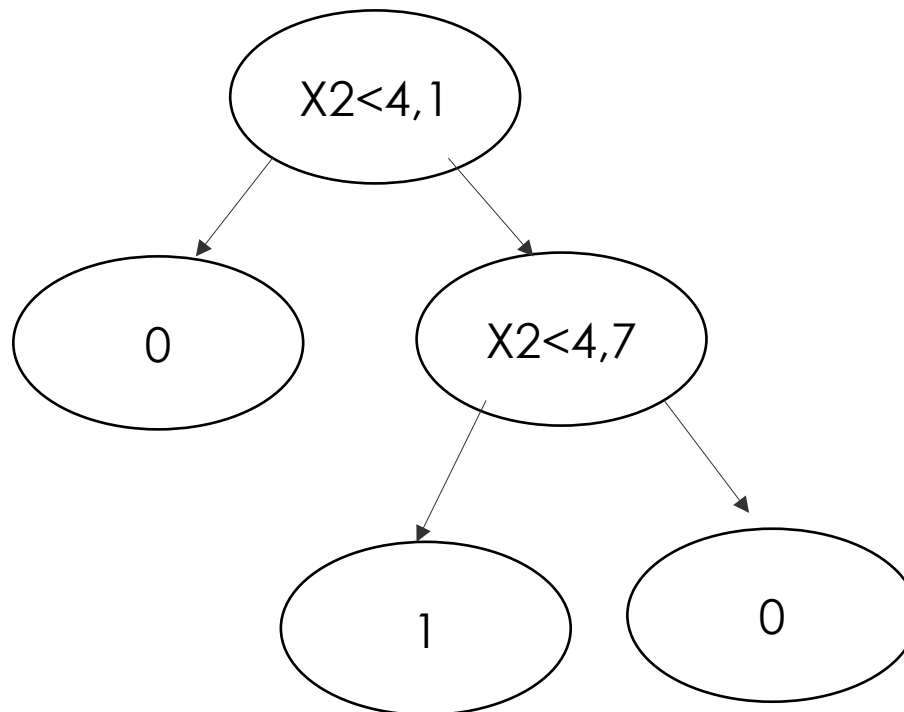
$9 > 7$, So we assume that the erroneous values are the 1

The final tree is :



Random Forest

Let's take the example of trees generated from the same dataset



Random Forest

- A very popular and powerful method
- It is less sensitive to training data
- Uses multiple decision trees
- Based on the principle of bagging

Random Forest

It is mainly based on two ideas: bootstrapping and aggregation:

Bootstrapping: allows the generation of random data from our original dataset. The process is as follows:

- Select a random number of rows.
- Perform random sampling with replacement.

Random Forest: Bootstrapping

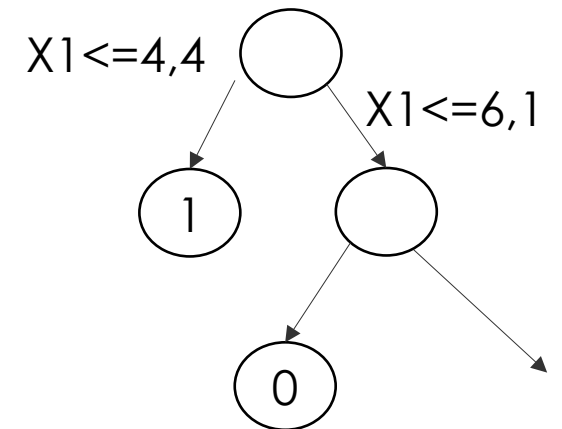
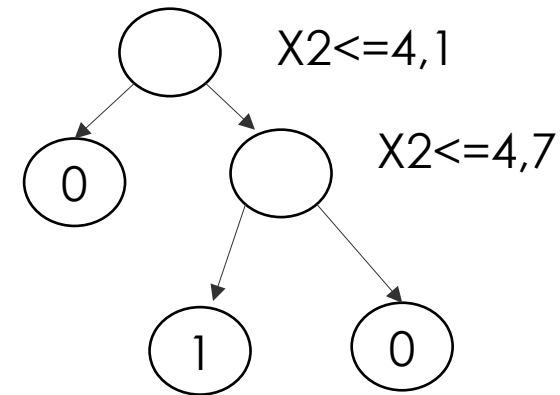
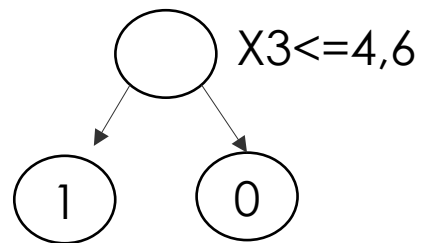
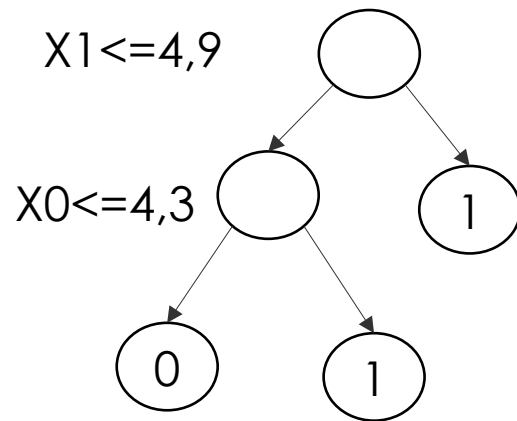
Id	x1	x2	x3	x4	x5	y
0	4,3	4,9	4,1	4,7	5,5	0
1	3,9	6,1	5,9	5,5	5,9	0
2	2,7	4,8	4,1	5	5,6	0
3	6,6	4,4	4,5	3,9	5,9	1
4	6,5	2,9	4,7	4,6	6,1	1
5	2,7	6,7	4,2	5,3	4,8	1

Id	Id	Id	Id
2	2	4	3
0	1	1	3
2	3	3	2
4	1	0	5
5	4	0	1
5	4	2	2

Random Forest: Bootstrapping

Let's take an example of data with 5 dimensions.

After bootstrapping, we will create a decision tree for each generated data point



Random Forest: Aggregation

How are we going to use this algorithm to classify objects?

Given a new data point:

2,8	6,2	4,3	5,3	5,5
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We will pass the object through each tree:

- Arbre 1: 1
- Arbre 2: 0
- Arbre 3: 1
- Arbre 4: 1

Random Forest

- The process of combining results from multiple models is called aggregation.
- Bootstrapping + aggregation = Bagging.
- It is a random forest since two random processes were used: data generation and feature selection. The optimal number of selected features: researchers have found that the log and sqrt of the total number of features are a good choice.

Random Forest

Advantages:

- Can handle large volumes of data with high dimensionality.
- Generally avoids overfitting.
- Provides excellent results for classification.

Disadvantages:

- While it excels in classification, it may not perform as well in regression.
- There is not much control over the model; multiple tests are needed to achieve good results.

Naïve bayes

- The statistical approach to classification formalizes the concept of class through probabilistic models.
- The n observations $x_1, x_2, \dots, x_n$ represent the random variable X .
- The labels $y_1, y_2, \dots, y_n$ represent the random variable y .
- The probability that an observation belongs to a class is determined by:
 - **An inference problem:** involves determining the probability distributions.
 - **A prediction problem:** involves using these distributions to predict the class of an observation.

Naïve bayes: Inference and prediction

- Given a set of features $x = (x_1, x_2, \dots, x_n)$. For prediction, we will consider these variables (X and Y) as random.
- To find the label y , we need to evaluate this expression for all possible values of y .
- The expression means, what is the probability that Y equals c given X equals $(x_1, x_2, \dots, x_n)$. Therefore, finding y aims to find the value of y that maximizes this probability.

Bayes' theorem

- Probabilistic reasoning heavily relies on Bayes' theorem, which allows us to express the conditional distribution

$$p(y = c|x) = \frac{p(x|y = c)p(y = c)}{p(x)}$$

- $p(y=c)$: is the prior distribution of labels before observing the data.
- $p(x|y=c)$: It is the likelihood, quantifying how likely we observe the realization of x from X given $y = c$.

Bayes' theorem

- $p(y=c | x)$: It is the posterior distribution of labels after observing the data.
- $p(x)$: is the marginal probability that x is observed
- For comparing probabilities, we can ignore the denominator

Bayes' theorem: Example

y	0	1	1	1	0	0	1	0	0	0
x1	0	0	1	0	2	1	0	2	2	1
x2	0	1	2	0	2	1	2	0	1	0

Estimate y for $x=(0,2)$

$$P(y=0) = 6/10 = 0,6$$

$$P(y=1) = 4/10 = 0,4$$

$$P(x=(0,2) | y=1) = 1/4 * p(y) = 1/10$$

$$P(x=(0,2) | y=0) = 0$$

Naïf bayes classifieur

The naive Bayes assumes that the variables x_i are independent

$$p(y = c | X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \frac{\prod_{j=1}^p p(X_j = x_j | y = c) p(y = c)}{p(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)}$$

- To calculate $p(x | y)$:

Categorical variables:

$$p(x_j | y) = \frac{\text{le nombre de types de la classe } y \text{ qui ont la valeur } x_j}{\text{le nombre des types de la classe } y}$$

Continuous variables: An attribute with a continuous value is generally assumed to have a Gaussian distribution with a standard deviation σ and a mean μ

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{\pi}\sigma} \exp \left(-\frac{(x-\mu)^2}{2\sigma^2} \right)$$
$$p(x_k | c_i) = g(x_k, \mu_{ci}, \sigma_{ci})$$

Naif bayes classifier

y	0	1	1	1	0	0	1	0	0	0
x1	0	0	1	0	2	1	0	2	2	1
x2	0	1	2	0	2	1	2	0	1	0

- Estimate y for (x,2) ?
- $P(x=(0,2) | y=1) = p(x1=0 | y=1) * p(x2=2 | y=1)$
 $= 3/4 * 2/4$
 $= 6/16$
- $P(x=(0,2) | y=0) = p(x1=0 | y=0) * p(x2=2 | y=0)$
 $= 1/6 * 1/6$
 $= 2/36$

Naif bayes classifier

Advantages:

- Simple and quick to implement
- Uses a small amount of data to estimate parameters
- Produces good results in many cases

Disadvantages:

- Assumes that variables are independent"